

Chaos on a Strange Attractor

Strogatz ch 9.3

→ Lorenz (1963) delivered a system of equations to represent weather behavior

→ After many changes of coordinates / manipulation, you arrive at

'Typical' choice for parameters

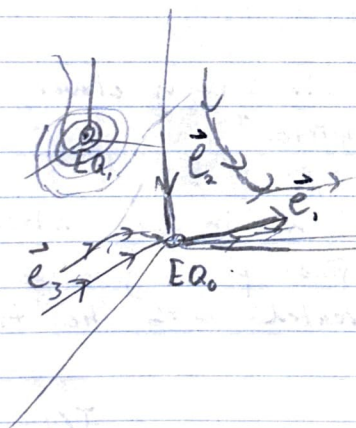
Lorenz Attractor

$$\begin{cases} \dot{x} = \sigma(y-x) \\ \dot{y} = rx - y - xz \\ \dot{z} = xy - bz \end{cases}$$

$\sigma = 10, b = 8/3, r = 28$

→ consider the equilibria, given σ, b, r above
• one is clearly $x=y=z=0$

origin
 $EQ_0: (x_1, x_2, x_3) = (12, -2.7, -23)$



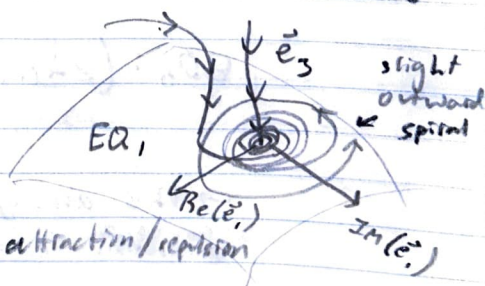
$EQ_1: (\mu \pm i\omega, \lambda^{(3)})$
 $= (0.1 \pm 10i, -14)$

↑
quickly
contracting

↑
dominant 'rotational'
component of the
motion

→ very small "pushing out"

→ What do the values of the eigenvalues communicate about the degree / timescale of attraction/repulsion



slight
outward
spiral

→ Consider EQ, : $(\mu \pm i\omega, \lambda) = (0.1 \pm i0i, -11)$

• Rotation period $2\pi/\omega \Rightarrow T_{eq} = 0.6$

→ rate exponent :

→ Multiplier : in one 'return' / time scale,
how far is the trajectory scaled in each
direction

$$\text{Multiplier} \rightarrow \Delta_{(1)} = e^{\mu^{(1)} T_{eq}}$$

rotation period
rate exponent

$$\text{EQ}_1 : (\Delta_{(1)}, \Delta_{(2)}, \Delta_{(3)}) = (1.1, 1.1, 2 \cdot 10^{-4})$$

→ "I have a friend who is very clever, computing systems
to arbitrary precision"

↳ If you zoom in in the attractive direction
($\vec{e}_3$) for EQ, fixed point, you see a larger-by-layer
fractal being created with the trajectory

$$\text{EQ}_0 : (\Delta_{(1)}, \Delta_{(2)}, \Delta_{(3)}) = (e^{T/2}, e^{-3E}, e^{-23T})$$

→ This analysis extends to N clearly dominant multiplier
dimensional systems

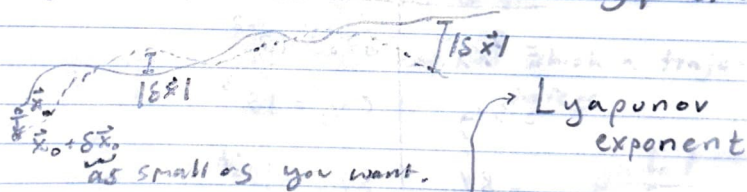
→ General strategy:

- compute equilibria
- determine their eigen-decomposition according to the linearization
- compare the 'degree' of dynamics using Lyapunov exponents / multipliers
- order directions / dimensions that are most - least important; they tend to arrange themselves exponentially

→ Stability: crucial to understand the difference between local & global stability;

Local Instability

- Consider an arbitrary system (metric space) with two adjacent initial conditions separated by $|\delta \vec{x}_0| \ll 1$



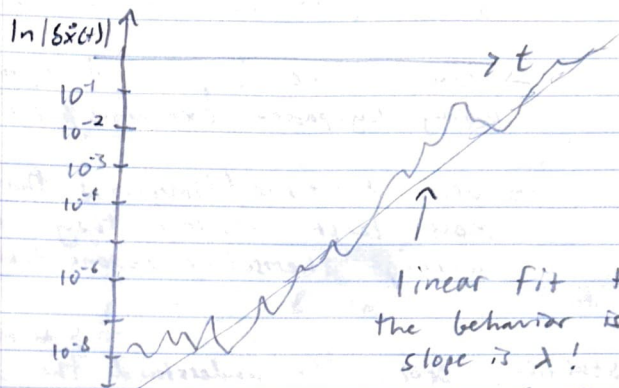
$$|\delta \vec{x}(t)| \approx |\delta \vec{x}_0| e^{\lambda t}$$

approximately / on average

$$|\delta \vec{x}(t)| = |\delta \vec{x}_0| e^{\lambda t}$$

$$\lambda = \lim_{t \rightarrow 0} \frac{1}{t} \ln \left| \frac{\delta \vec{x}(t)}{\delta \vec{x}_0} \right|$$

Lyapunov
Exponent



→ but this doesn't always work

$$t_{\text{lyap}} = t_{\text{horizon}} = - \frac{\ln |\delta \vec{x}(t)|}{\lambda}$$

t_{lyap}	
$= 2/\lambda$	$\delta x_0 = 10^{-2}$
$= 3/\lambda$	$\delta x_0 = 10^{-3}$
$= 8/\lambda$	$\delta x_0 = 10^{-8}$

You cannot beat exponentials

→ if your system is exponentially unstable, no amount of finite precision will yield an indefinitely stable trajectory

→ Chaos $\neq$ unstable

↓
Chaos means 'you are condemned forever to spiral around & never settle down'

→ unstable can mean local instability that is 'resolved' quickly

↓
Local instability + Global Recurrence

↳ you need a 'choice' between unstable eqilibria, like EA_1 & EA_2 in the Lorenz system

→ at every point in time, you have a finite number of 'global' choices.

→ this 'number of choices' can be computed, but is difficult...

→ we define the log of # of ways to return

Entropy: $\lim_{t \rightarrow \infty} \left(\ln(\# \text{ returns}) / t \right) \equiv h$

↳ the rate at which a trajectory makes its choices

local

→ this is unstable, trajectory leaves

global

→ there are many ways to come back

(a coarse def):

→ Chaos is positive Lyapunov + positive Entropy

this is what makes it hard...